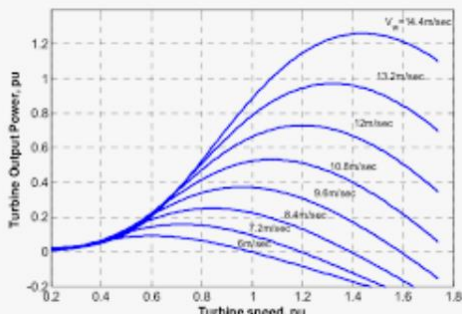
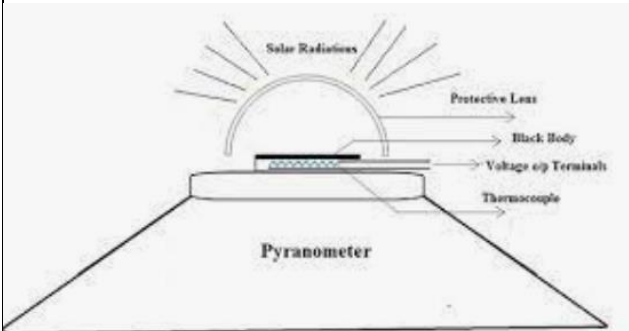
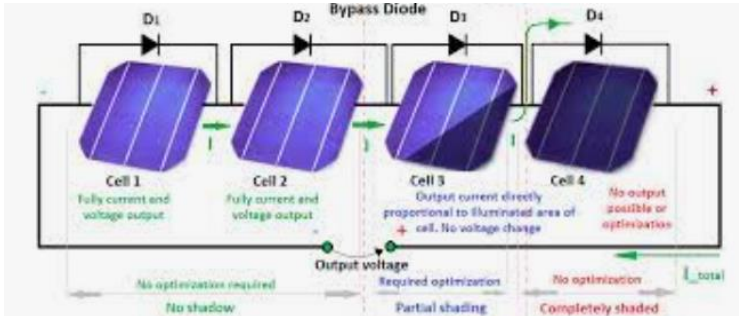
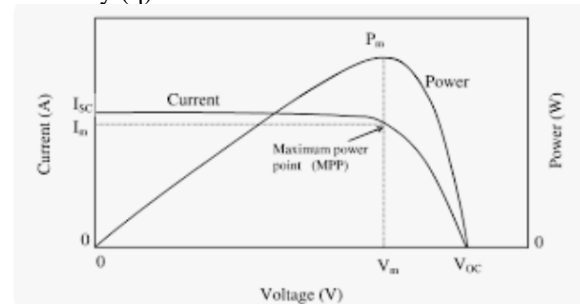
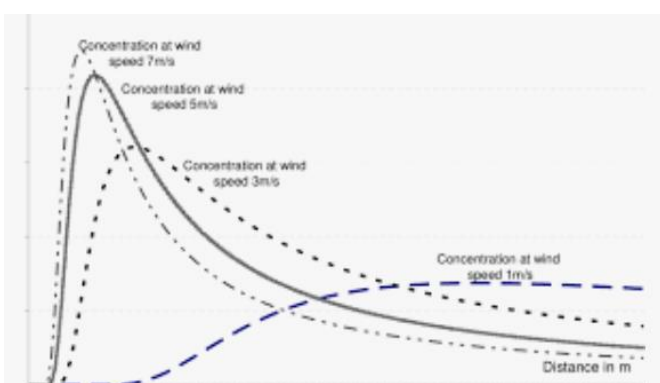


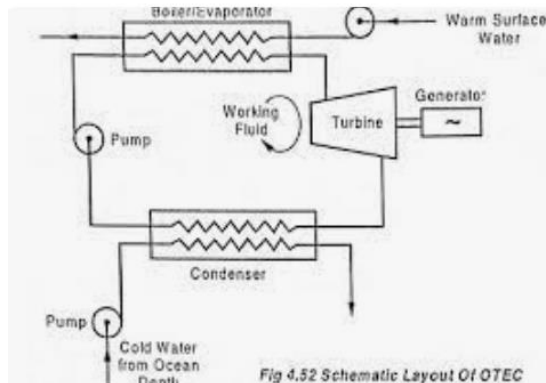
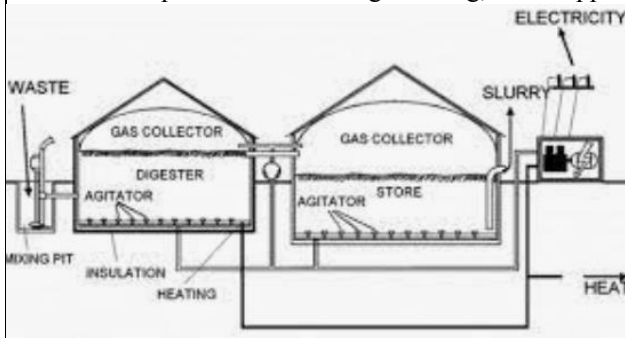
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY																																	
SIXTH SEMESTER B.TECH DEGREE (R,S)EXAMINATION, June 2023(2019 Scheme)																																	
Course Code: EET322																																	
Course Name: RENEWABLE ENERGY SYSTEMS																																	
Max. Marks: 100			Duration: 3 Hours																														
PART A																																	
		Answer all questions, each carries 3 marks.	Marks																														
1		Global warming is an aspect of climate change, referring to the long-term rise of the planet's temperatures. It is caused by increased concentrations of greenhouse gases in the atmosphere, mainly from human activities such as burning fossil fuels, and farming.	(3)																														
2		It causes respiratory diseases in humans. It causes global warming. Pollutants like oxides of sulfur and nitrogen when getting dissolved in rainwater and cause acid rain. The release of certain chemicals causes depletion of the ozone layer.	(3)																														
3		Insolation is the amount of solar radiation received on a given surface in a given time period. In particular, daily insolation Q ($\text{MJ m}^{-2} \text{ day}^{-1}$) is the solar radiation incident on a horizontal surface per square metre integrated over a day.	(3)																														
4		In this method, the optimum tilt angle for solar panels during winter is calculated by multiplying the latitude by 0.9 and then adding 29° . In the above case example of a latitude of 34° , the tilt angle will be $(34 * 0.9) + 29 = 59.6^\circ$.	(3)																														
5		<table><tr><th>Factor</th><th>VAWT</th><th>HAWT</th></tr><tr><td>Wind direction</td><td>Omni-directional</td><td>Head on</td></tr><tr><td>Wind conditions</td><td>Varying</td><td>Consistent</td></tr><tr><td>Machinery location</td><td>Base</td><td>Tower top</td></tr><tr><td>External aid</td><td>Yes</td><td>No</td></tr><tr><td>Blade</td><td>Simple to complex</td><td>Detailed and complex</td></tr><tr><td>Forces</td><td>Tension, fatigue and bending moments</td><td>Root bending moments, fatigue</td></tr><tr><td>Torque</td><td>Sinusoidal</td><td>Constant</td></tr><tr><td>Noise</td><td>Quiet</td><td>Noticeable</td></tr><tr><td>Energy conversion effectiveness</td><td>$\lll$ Betz limit</td><td>75-80% of Betz limit</td></tr></table>	Factor	VAWT	HAWT	Wind direction	Omni-directional	Head on	Wind conditions	Varying	Consistent	Machinery location	Base	Tower top	External aid	Yes	No	Blade	Simple to complex	Detailed and complex	Forces	Tension, fatigue and bending moments	Root bending moments, fatigue	Torque	Sinusoidal	Constant	Noise	Quiet	Noticeable	Energy conversion effectiveness	$\lll$ Betz limit	75-80% of Betz limit	(3)
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6			(3)
7		Tides are easily predictable. Inexpensive to maintain. Reliable and renewable source of energy. High energy density than other renewable energy forms. It produces no greenhouse gases or other waste. Water is hundreds of times denser than air, which makes tidal energy more powerful than wind. It is more efficient than wind or solar energy due to its relative density and produces no greenhouse gases or other waste, making it an attractive renewable energy source to pursue.	(3)
8		Biofouling is referred to as the unwanted de- position and growth of biofilms. This phenomenon can occur in an extremely wide range of situations, from the colonisation of medical devices to the production of ul- tra-pure, drinking and process water and the fouling of ship hulls, pipelines and reservoirs.	(3)
9		The three principal methods of thermochemical conversion of MSW are combustion (in excess air), gasification (in reduced air), and pyrolysis (in absence of air). The most common technique for producing both heat and electrical energy from wastes is direct combustion	(3)
10		Fuel cells work like batteries, but they do not run down or need recharging. They produce electricity and heat as long as fuel is supplied. A fuel cell consists of two electrodes—a negative electrode (or anode) and a positive electrode (or cathode)—sandwiched around an electrolyte.	(3)
PART B			
<i>Answer one full question from each module, each carries 14 marks.</i>			
		Module I	
11	a)	Energy generated by using wind, solar, small hydro, tides, geothermal heat and biomass is known a non-conventional energy. All these sources are renewable process of energy generation and do not cause environmental pollution.	(4)
	b)	1) A Fuel Supply That Never Runs Out. ... 2) Zero Carbon Emissions. ... 3) Cleaner Air and Water. ... 4) A Cheaper Form of Electricity. ... 5) Renewable Energy Creates New Jobs. ... 1) Higher Capital Costs. ... 2) Electricity Production Can Be Unreliable. ...	(10)

		3) Energy Storage Is a Challenge.	
		OR	
12	a)	Green energy is important for the environment as it replaces the negative effects of fossil fuels with more environmentally-friendly alternatives. Derived from natural resources, green energy is also often renewable and clean, meaning that they emit no or few greenhouse gases and are often readily available	(4)
	b)	Energy resources are all forms of fuels used in the modern world, either for heating, generation of electrical energy, or for other forms of energy conversion processes. Energy resources can be roughly classified in three categories: renewable, fossil, and nuclear.	(10)
		Module II	
13	a)	Linear Concentrator Systems. Linear concentrator systems collect the sun's energy using long rectangular, curved (U-shaped) mirrors. ... Dish/Engine Systems. ... Power Tower Systems. ... Additional Resources.	(9)
	b)	A pyranometer is a sensor that converts the global solar radiation it receives into an electrical signal that can be measured. Pyranometers measure a portion of the solar spectrum. As an example, the CMP21 Pyranometer measures wavelengths from 0.285 to 2.8 μm	(5)
			
		OR	

14	a)	Under partial shading conditions, photovoltaic (PV) arrays are subjected to different irradiance levels caused by nonuniform shading. As a result, a mismatch between the modules, a reduction in the power generated, and the hotspot phenomenon will be observed.  The diagram illustrates a PV array with four cells (Cell 1, Cell 2, Cell 3, Cell 4) connected in series. Each cell has a bypass diode (D1, D2, D3, D4) connected in parallel. Cell 1 and Cell 2 are fully illuminated, Cell 3 is partially shaded, and Cell 4 is completely shaded. The output voltage is shown as the sum of the voltages of the cells. The output current is shown as the minimum of the currents of the cells. The diagram also shows the power output of the array, which is reduced due to the shading of Cell 3 and Cell 4.	(3)
	b)	The basic characteristics of a solar cell are the short-circuit current (ISC), the open-circuit voltage (VOC), the fill factor (FF) and the solar energy conversion efficiency (η).  The graph shows the current (A) and power (W) characteristics of a solar cell. The current curve starts at I_{sc} (short-circuit current) and decreases to zero at V_{oc} (open-circuit voltage). The power curve starts at zero, reaches a maximum at P_m (maximum power), and then decreases to zero at V_{oc}. The maximum power point (MPP) is marked on the power curve.	(11)
		Module III	
15	a)	Wind speed affects weather forecasting, aviation and maritime operations, construction projects, growth and metabolism rate of many plant species, and has countless other implications.  The graph shows the concentration of a substance versus distance in meters for different wind speeds. The concentration increases with distance and then decreases. The concentration is highest for a wind speed of 7 m/s and lowest for a wind speed of 1 m/s.	(6)
	b)	There are two main types of hydropower turbines: reaction and impulse. The type of hydropower turbine selected for a project is based on the height of standing water—referred to as "head"—and the flow, or volume of water over time, at the site.	(8)
		OR	

16	a)		(4)
	b)	Derive the expression for maximum power extracted by a wind turbine...10 marks	(10)
		Module IV	
17	a)	Describe the principle...2 marks, various components 2 marks.	(4)
	b)		(10)
		OR	
18	a)	High Initial Cost. We can conclude that harnessing OTEC involves expensive equipment that must be purchased and installed. ... Locality of Production. ... Small Temperature Difference. ... Harmful Effects on Marine Life. ... Interferes with Navigation. ... Large Size Turbines with Expensive Liquid.	(4)
	b)		(10)

		 Fig 4.52 Schematic Layout Of OTEC	
		Module V	
19	a)	Poor collection, improper segregation, a lack of vehicles and inadequate waste transportation increase the risk of supply chain disruption and create a barrier for utilising waste in biogas production.	(4)
	b)	balloon plants, - fixed-dome plants, - floating-drum plants. A balloon plant consists of a plastic or rubber digester bag, in the upper part of which the gas is	(10)
			
		OR	
20	a)	They are essential in modern smart grids, meeting changing energy demands, such as electric mobility. Energy storage provides flexibility and opportunities for remote areas using various technologies, including electro-mechanical, chemical, thermal, and electrochemical (batteries).	(4)
	b)	Some forms of storage that produce electricity include pumped-storage hydroelectric dams, rechargeable batteries, thermal storage including molten salts which can efficiently store and release very large quantities of heat energy, and compressed air energy storage, flywheels, cryogenic systems and superconducting	(10)
